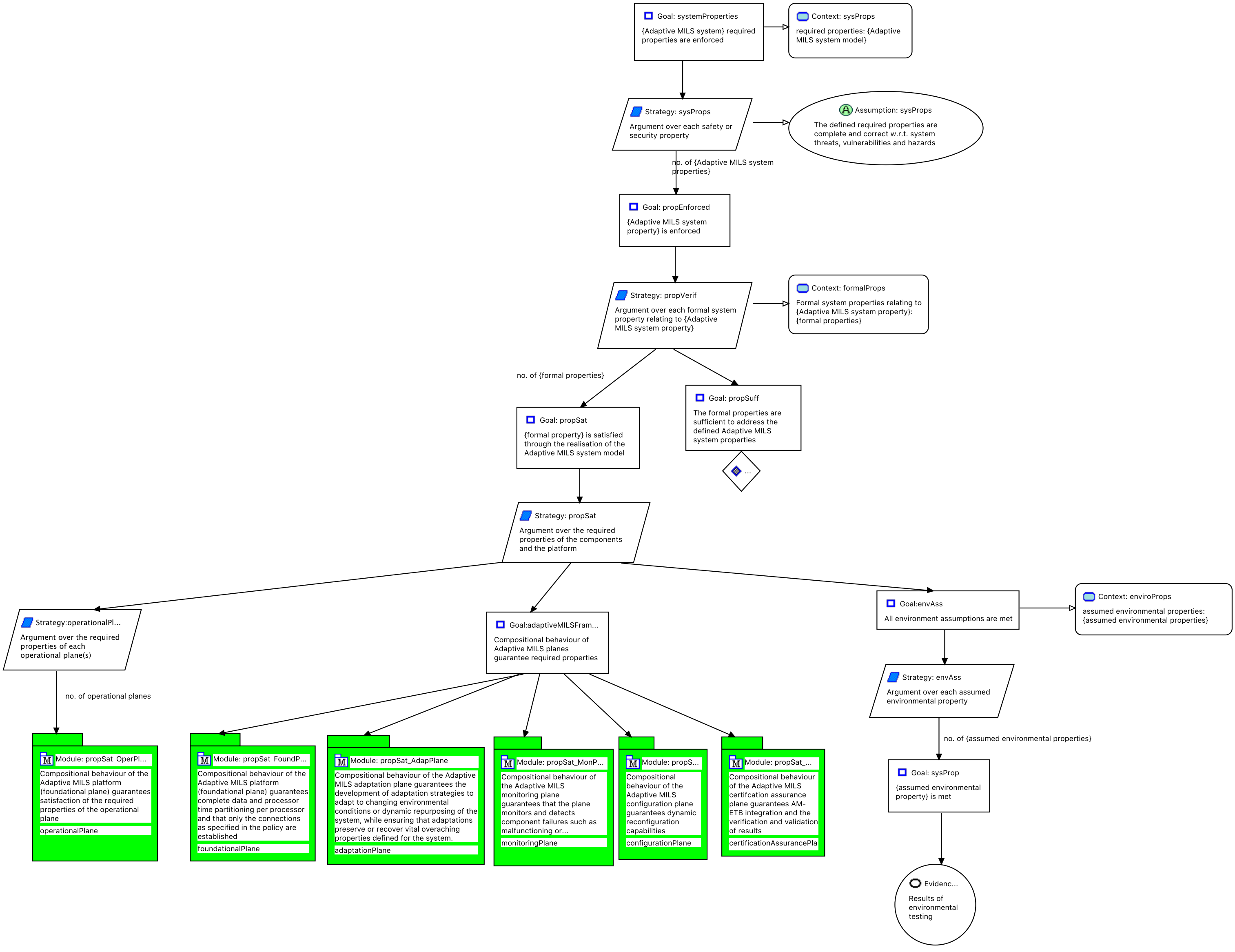
Goal: systemProperties
{Adaptive MILS system} required properties are enforced
Context: sysProps
required properties: {Adaptive MILS system model}
Strategy: sysProps
Argument over each safety or security property
Assumption: sysProps
The defined required properties are complete and correct w.r.t. system threats, vulnerabilities and hazards
no. of {Adaptive MILS system properties}
Goal: propEnforced
{Adaptive MILS system property} is enforced
Strategy: propVerif
Argument over each formal system property relating to {Adaptive MILS system property}
Context: formalProps
Formal system properties relating to {Adaptive MILS system property}: {formal properties}
no. of {formal properties}
Goal: propSat
{formal property} is satisfied through the realisation of the Adaptive MILS system model
Goal: propSuff
The formal properties are sufficient to address the defined Adaptive MILS system properties
...
Strategy: propSat
Argument over the required properties of the components and the platform
Strategy:operationalPl...
Argument over the required properties of each operational plane(s)
Goal:adaptiveMILSFram...
Compositional behaviour of Adaptive MILS planes guarantee required properties
Goal:envAss
All environment assumptions are met
Context: enviroProps
assumed environmental properties: {assumed environmental properties}
Strategy: envAss
Argument over each assumed environmental property
no. of {assumed environmental properties}
Goal: sysProp
{assumed environmental property} is met
Evidenc...
Results of environmental testing
no. of operational planes
Module: propSat_OperPl...
Compositional behaviour of the Adaptive MILS platform (foundational plane) guarantees satisfaction of the required properties of the operational plane
operationalPlane
Module: propSat_FoundP...
Compositional behaviour of the Adaptive MILS platform (foundational plane) guarantees complete data and processor time partitioning per processor and that only the connections as specified in the policy are established
foundationalPlane
Module: propSat_AdapPlane
Compositional behaviour of the Adaptive MILS adaptation plane guarantees the development of adaptation strategies to adapt to changing environmental conditions or dynamic repurposing of the system, while ensuring that adaptations preserve or recover vital overaching properties defined for the system.
adaptationPlane
Module: propSat_MonP...
Compositional behaviour of the Adaptive MILS monitoring plane guarantees that the plane monitors and detects component failures such as malfunctioning or...
monitoringPlane
Module: propS...
Compositional behaviour of the Adaptive MILS configuration plane guarantees dynamic reconfiguration capabilities
configurationPlane
Module: propSat_...
Compositional behaviour of the Adaptive MILS certifcation assurance plane guarantees AM-ETB integration and the verification and validation of results
certificationAssurancePla